Explaining Transaction-Level Sales with OLS Regression and Principal Component Analysis

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Abstract. This report analyses 2,823 commercial transactions to identify the drivers of sales revenue (SALES). An ordinary least squares (OLS) model of SALES on quantity ordered, unit price and manufacturer's suggested retail price (MSRP) explains 79.6% of the variance, but its residuals are strongly non-normal and heteroscedastic. Variance inflation factors (VIF, maximum 1.82) show that multicollinearity is only moderate, so a principal component regression (PCR) brings no gain in fit (R^2 = 0.794). A log-log specification with HC3 robust standard errors reaches R^2 = 0.906 and gives elasticities close to one for quantity and price, which largely reflects the arithmetic structure of revenue. Quantity and price are statistically independent, so the data do not identify a demand elasticity.

1. Data and objective

The data come from the public Kaggle dataset *Sample Sales Data* (2,823 rows, 25 columns). Four quantitative variables are retained: QUANTITYORDERED, PRICEEACH, MSRP and SALES, together with four categorical variables (PRODUCTLINE, DEALSIZE, COUNTRY, STATUS) reserved for the later correspondence analysis and classification stages. The objective is to quantify the effect of the controllable levers (price and volume) on revenue using OLS, and to examine the structure of the quantitative space using principal component analysis (PCA), implemented with scikit-learn and statsmodels in Python.

2. Preparation and exploratory analysis

No missing values were found in the eight selected columns, so the clean sample keeps all 2,823 observations. A Mahalanobis-distance screen on the three explanatory variables (threshold: chi-square, 3 d.f., $p < 0.001$) flags only 13 multivariate outliers (0.46%), which were kept. Table 1 reports the univariate statistics. SALES is right-skewed (1.16) and PRICEEACH is left-skewed (-0.95), which anticipates departures from normality in the regression residuals.

Table 1. Descriptive statistics of the quantitative variables (n = 2,823).

Variable	Mean	Std. dev.	Median	Min	Max	Skewness
QUANTITYORDERED	35.09	9.74	35.0	6.00	97.0	0.363
PRICEEACH	83.66	20.17	95.7	26.88	100.0	-0.947
MSRP	100.72	40.19	99.0	33.00	214.0	0.580
SALES	3553.89	1841.87	3184.8	482.13	14082.8	1.161

Among the categorical variables, Classic Cars (34.25%) and Vintage Cars (21.50%) dominate PRODUCTLINE; Medium (49.03%) and Small (45.41%) deals dominate DEALSIZE; the USA accounts for 35.57% of orders; and 92.70% of orders are Shipped. Six countries and four status levels fall below the 2% frequency threshold and should be grouped before any correspondence analysis. A chi-square test of independence between PRODUCTLINE and DEALSIZE rejects independence (chi^2 = 135.36, 12 d.f., $p = 5.2 \times 10^{-23}$), but Cramér's V is only 0.155, a weak association; the test should be read with care because some cells are empty or very small (for example, no Large deals for Ships).

Pearson correlations show that QUANTITYORDERED is essentially uncorrelated with price (r = 0.006) and with MSRP (r = 0.018), whereas PRICEEACH and MSRP are positively correlated (r = 0.671). SALES correlates with quantity (0.551), price (0.658) and MSRP (0.635). The correlation between SALES and the product QUANTITYORDERED x PRICEEACH is 0.864, so revenue is strongly, but not exactly, the product of volume and price in this file.

3. Ordinary least squares

The baseline model is $SALES_i = b_0 + b_1\ QUANTITYORDERED_i + b_2\ PRICEEACH_i + b_3\ MSRP_i + e_i$. The model is globally significant ($F(3, 2819) = 3665$, $p < 0.001$) with R^2 = adjusted $R^2 = 0.796$. Each additional unit ordered raises revenue by about 102.7 currency units, each additional unit of price by about 39.0, and each unit of MSRP by about 15.5, all other things equal. All coefficients are significant at the 1% level (Table 2).

Table 2. OLS estimates for SALES (all p-values < 0.001).

Term	Coefficient	Std. error	t	HC3 std. error	VIF
Intercept	-4878.88	87.54	-55.74	106.19	-
QUANTITYORDERED	102.67	1.61	63.81	2.03	1.00
PRICEEACH	39.04	1.05	37.28	1.51	1.82
MSRP	15.52	0.53	29.53	0.95	1.82

Multicollinearity. The largest VIF is 1.82, well below the usual alert thresholds of 5 or 10. The correlation between PRICEEACH and MSRP is therefore noticeable but does not destabilise the estimates. The condition number (789) mainly reflects the unstandardised scales of the variables. **Residual diagnostics.** The Jarque-Bera statistic is 10,245.7 (skewness 1.95, kurtosis 11.48), so normality of the errors is rejected. The Breusch-Pagan test also rejects homoscedasticity ($p = 9.3 \times 10^{-30}$). The Durbin-Watson statistic is 1.578, but the rows are not an ordered time series, so this value has no clear interpretation here. The Gauss-Markov conditions on error variance are thus violated, and the classical standard errors and confidence intervals are unreliable.

Robust and log-log specifications. HC3 heteroscedasticity-robust errors leave the coefficients unchanged and increase the standard errors by 21% to 81%; every effect remains significant. A log-log model (Table 3) improves the fit to $R^2 = 0.906$ and reduces kurtosis from 11.48 to 5.88, although normality is still rejected (Jarque-Bera = 1,717.8). Its coefficients are elasticities: a 1% increase in quantity is associated with a 1.00% increase in sales, a 1% increase in price with 1.06%, and a 1% increase in MSRP with 0.28%, holding the others constant.

Table 3. Log-log model for SALES ($R^2 = 0.906$, Durbin-Watson = 1.178).

Term	Coefficient	HC3 std. error	z	95% CI
Intercept	-1.371	0.056	-24.44	[-1.481, -1.261]
log QUANTITYORDERED	0.9999	0.010	96.46	[0.980, 1.020]
log PRICEEACH	1.0577	0.023	45.45	[1.012, 1.103]
log MSRP	0.2773	0.019	14.58	[0.240, 0.315]

4. Principal component analysis and PCR

PCA was applied to the standardised explanatory variables (Table 4). Two components have eigenvalues of at least 1 and retain 89.02% of the total inertia. The second eigenvalue (1.0001) only marginally passes the Kaiser rule: because QUANTITYORDERED is almost uncorrelated with the other variables, it forms its own axis with an eigenvalue close to 1 by construction.

Table 4. Eigenvalues and inertia of the PCA on the three explanatory variables.

Axis	Eigenvalue	Inertia (%)	Cumulative (%)	Kaiser rule
PC1	1.6716	55.70	55.70	Retained
PC2	1.0001	33.32	89.02	Retained (marginal)
PC3	0.3294	10.98	100.00	Discarded

The loadings give a clear reading. PC1 weights PRICEEACH and MSRP equally (0.707 each) and is a *unit value* axis. PC2 is almost exactly QUANTITYORDERED (1.000) and is a *volume* axis. PC3 contrasts PRICEEACH (-0.707) with MSRP (+0.707), that is, the transaction price relative to the list price, which can be read as discount depth. This discarded axis carries 10.98% of the inertia and some explanatory information.

Regressing SALES on PC1 and PC2 (PCR) gives R^2 = 0.794 with coefficients of 1022.3 (PC1) and 973.4 (PC2), both significant (t = 84.0 and 61.9), and an intercept equal to the mean of SALES (3553.9). All VIFs equal 1.0, but this follows from orthogonality by construction and is not evidence of a better model. The PCR is slightly worse than the raw OLS model (AIC 46,001.6 versus 45,981.5), because the discarded PC3 contains useful information, and it inherits the same non-normal residuals (Jarque-Bera = 11,123.1).

A second PCA on the four standardised quantitative variables, used for the graphical representation, yields a first plane that explains 88.66% of the inertia (59.82% + 28.84%). Table 5 gives the variable coordinates. Axis 1 groups PRICEEACH, MSRP and SALES (value axis); Axis 2 is driven by QUANTITYORDERED (volume axis), and SALES, which depends on both, projects on both axes.

Table 5. Variable coordinates on the first factorial plane (4-variable PCA).

Variable	Axis 1 (59.82%)	Axis 2 (28.84%)
QUANTITYORDERED	0.380	0.910
PRICEEACH	0.838	-0.359
MSRP	0.831	-0.353
SALES	0.925	0.268

5. Model comparison

Table 6. Comparison of the three models.

Model	R^2	AIC	Max VIF	Main remark
OLS (raw)	0.796	45,981.5	1.82	Non-normal, heteroscedastic residuals
PCR (2 components)	0.794	46,001.6	1.00	No gain; loses discount information
Log-log (HC3)	0.906	-2,323*	1.82	Best fit; interpretable elasticities

* The AIC of the log-log model is not comparable with the others because its dependent variable is log(SALES).

6. Conclusion and recommendations

(i) Volume and unit price are statistically independent (r = 0.006) and each contributes strongly to revenue. Because price and quantity do not co-move, the regression describes how revenue is composed and does not estimate how customers react to a price change; elasticities of about one for quantity and price mainly mirror the fact that revenue is roughly price times quantity. (ii) Multicollinearity between PRICEEACH and MSRP is moderate (VIF 1.82), so PCR is not needed and slightly degrades the model. (iii) The preferred specification is the log-log model with HC3 errors, whose inference does not rely on homoscedasticity; non-normality remains, so prediction intervals should be treated with caution, although the large sample (n = 2,823) supports the asymptotic tests. (iv) For management, price and volume can be steered as separate levers, and the gap between the transaction price and MSRP (discounting) deserves its own monitoring. **Limitations.** The data are cross-sectional with no usable time order, SALES is partly a function of the regressors, and the results describe association, not causation. The categorical structure (correspondence analysis) and the predictive classification (SVM) are treated in the next stages of the project.

Data and software sources

Kyanyoga. *Sample Sales Data*. Kaggle. https://www.kaggle.com/datasets/kyanyoga/sample-sales-data

scikit-learn developers. *scikit-learn: Machine Learning in Python*. GitHub repository. https://github.com/scikit-learn/scikit-learn